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Understanding the beliefs and attitudes towards mental health problems held by Muslim communities and acceptability of Cognitive Behavioural Therapy as a treatment: Systematic Review and Thematic Synthesis Protocol

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Understanding the beliefs and attitudes towards mental health problems held by Muslim communities and acceptability of Cognitive Behavioural Therapy as a treatment: Systematic Review and Thematic Synthesis Protocol

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Authors Contributions HaH drafted the manuscript and led the design of the protocol and provided input across all areas. PF provided guidance during development of the protocol, provided advice across all area and commented on all drafts. KL provided comments on full drafts.

Word count: 1932

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Abstract

Introduction Many beliefs regarding mental health difficulties alongside attitudes toward psychotherapy and Cognitive-Behaviour Therapy (CBT) in particular are held by people within Muslim communities. The aim of this systematic review is to gain an understanding regarding these beliefs and attitudes and consider how these vary across different Muslim communities.

Methods and Analysis Systematic review of PubMed/MEDLINE, CINAHL, PsycINFO, Ovid MEDLINE, Embase and Index Islamicus databases for studies published in English in addition to the Saudi Digital Library (SDL) for studies published in Arabic. The review will include quantitative, qualitative, and mixed methods studies published in peer-review journals since 1980. Methodological quality and risk of bias of included papers will be critically appraised independently according to the Mixed Methods Appraisal Tool (MMAT). Thematic synthesis will be used to extract outcome and analyse data from the included studies.

Ethics and Dissemination There are considered to be no ethical issues. Findings will be disseminated in both English and Arabic to clinicians and researchers via journal publication and conference presentation(s).

Trial Registration Number PROSPERO and registration number: CRD42020192854

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Strengths and limitations of this study

- This is the first systematic review and thematic synthesis examining the attitude and beliefs of Muslim communities toward mental health difficulties and Cognitive Behaviour Therapy as a treatment.
- The review is inclusive regarding attitudes and beliefs held by different Muslim communities to facilitate comparisons.
- The review includes a large number of participants drawn from different countries to facilitate a wider range of comparisons.
- Poor standard of reporting the methods within specific studies made it difficult to determine research quality.
- Transferability of results may be limited given that studies were restricted to those published in English or Arabic whilst many studies of the Muslim community are published in other languages.

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Introduction

Islam is the dominant religion in 56 countries around the globe and has more than 1.9 billion followers[1]. Most Muslims reside in the Islamic world that primarily covers the Middle East, Africa, and Asia. However, many Muslims can also be found in most other countries as a minority group having immigrated for economic, political or social reasons[2].

There are many beliefs held among Muslim communities concerning the aetiology of mental health difficulties, including Evil Eye, possession and black magic[3]. A predominant belief however is that life problems are interpreted as religious problems and this can lead to interpretation of psychological difficulties as evidence of a 'weakness of faith', and punishment from God (Allah) for sinful acts[4, 5, 6]. These beliefs are often linked and used to account for the experience of physical illness also[7]. Such interpretations as to the aetiology of mental health difficulties have a direct impact on members of the Muslim community seeking treatment for common mental health difficulties[3, 6]. Within England, this accounts for lower rates of help-seeking and recovery amongst members of the Muslim community[8].

Beliefs that mental health difficulties result from a 'weakness of faith' contributes to the most commonly adopted treatment having a religious basis[9]. This converts the understanding and experience of mental health difficulties into beliefs that they represent a test from God (Allah) because of the whispers of devil[10, 3]. As a consequence of these beliefs, treatment involves religious healers reading the Quran and praying for the recovery of the person until God is pleased and psychological problems disappear[11]. Such beliefs potentially result in psychological therapies, such as Cognitive Behavioural Therapy (CBT), being rejected by many in Muslim communities with a preference for traditional treatment. This represents a barrier to accessing services that offer psychological treatments for mental health difficulties

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such as anxiety and depression[12], with specific barriers identified for members of the Black Asian and Minority Ethnic (BAME) community[13].

A systematic review has identified a range of barriers and obstacles engaging with mental health interventions across a range of religious groups within Middle Eastern Arab countries[14]. Barriers included the acceptance of interference in the cultural context (e.g., beliefs and values and the stigma of mental illnesses), community and system factors (e.g., access to services) and clinical participation processes (e.g., the expectations of the client)[14]. These barriers are similar to those reported in Western societies that included access to effective treatment for anxiety and depression that included cultural, provider such as lack of knowledge of mental health problems, and systemic factors such as the limited availability of specialty mental health providers[12]. Somewhat surprisingly however, individual barriers[12] relating to an understanding of mental health and treatment were not reported.

Aims and Objectives

The systematic review aims to understand beliefs and attitudes held by different adult Muslim communities towards mental health problems, acceptability of a CBT model and treatment and appreciate barriers accessing CBT for this underserved population. These areas will be examined across the Muslim communities to gain an appreciation regarding the extent to which any differences permit conclusions to be generalised.

Objectives:

To gain an understanding of mental health beliefs held by different Muslim communities.

To appreciate beliefs and attitudes regarding causes and treatment of mental health difficulties.

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To evaluate the extent to which beliefs held towards mental health and treatment vary between Muslim communities.

To identify barriers accessing mental health treatment exist and vary between Muslim communities.

To appreciate the acceptability of CBT for the treatment of mental health difficulties held by different Muslim communities.

Method

This systematic review protocol adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA) (see online supplementary appendix 1).

Eligibility Criteria:

Empirical studies published in peer-review journals and meeting review-specific eligibility criteria informed by the SPIDER search tool for use in systematic reviews of qualitative and mixed method research[15]. All studies will be published since 1980 in the English and Arabic language. Grey literature will be excluded.

Study Population:

The study population includes working age male and female adult Muslims that have not received psychological therapies for the treatment of mental health difficulties as well as patients with experience of CBT treatment. Working age varies between countries and as such will differ as an inclusion criterion. On occasions participant religion may not be specified in the description of the participants. In such cases, studies conducted in countries where the population is above 90% Muslim (e. g., Afghanistan, Iran, Libya, Saudi Arabia,

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Somalia, Turkey, Pakistan) will be included. Studies that focus on health care professionals, family or carers views will be excluded.

Interventions:

Studies focussing on the understanding of mental health will not be limited to any specific psychological intervention. However, studies focused on acceptability towards psychoeducation and treatment approaches will be restricted to High and Low-intensity CBT[16].

Comparators:

Regardless of any comparator condition used in quantitative studies, all studies meeting inclusion criteria will be included.

Study Designs:

Qualitative, quantitative, questionnaire-based and mixed methods research designs will be included.

Outcomes:

Understanding of the beliefs and attitudes held towards the causes and treatment of mental health problems, appreciation regarding the acceptability of CBT for the understanding and treatment of mental health difficulties, identification of barriers accessing evidence-based psychological interventions for mental health difficulties.

Language:

Studies in the English or Arabic language.

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Patient and Public Involvement:

No patient involved

Information sources

In the English language the search will be based on the following research databases: PubMed/MEDLINE, CINAHL, PsycINFO, Ovid MEDLINE, Embase, and the Index Islamicus religious database. The Saudi Digital Library (SDL) electronic database will be searched for publications in Arabic.

Search Strategy

Truncation# or Wild Cards* will be used following appropriate search terms to support searches for variations on a word formed through different suffixes. For example, Depressi* to include depression or depressive disorder or depressive symptoms or major depressive disorder. Or # if the word has a different spelling such as behaviour#. Where available, search terms will be adjusted for use with specific bibliographic databases filters. Truncation or Wild Cards are not used within the SDL. Reference lists will be hand searched for all included studies to identify further studies (see online supplementary appendix 2, 3).

Study Selection

Prior to screening, a calibration exercise will be undertaken between HaH and PF. All titles and abstracts yielded by the search against inclusion/exclusion criteria will be screened by HaH. Full text of articles included from the title abstract screen will be independently reviewed by HaH and PF with discrepancies regarding inclusion resolved through discussion.

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Where consensus is not reached, differences will be resolved by wider discussion with KL for final decision making.

Data Extraction

Descriptive data regarding specific background characteristics –population, year of publication, country of study, study aim, study design, population, and type of mental health difficulties if experienced– will be extracted from each study included in the review. Consistent with the aim of qualitative synthesis to extend beyond primary studies, additional data extracted will focus on the analysis and interpretations of the study authors towards themes, sub-themes, beliefs, models, attitudes and other relevant information associated with the study objectives. Data extraction forms will be developed and piloted a priori. For English studies data extraction will be carried out by HaH with a random sample checked by PF. Data extraction for studies published in the Arabic language will only be undertaken by HaH.

Risk of Bias of Individual Studies

Methodological quality and risk of bias of included papers will be critically appraised independently by HaH and PF according to the Mixed Methods Appraisal Tool (MMAT)[17].

Data Analysis and Synthesis

Thematic synthesis[18] will be used to support interpretation of data obtained from qualitative, quantitative and mixed methods research beyond that undertaken by authors of included studies. Arising data will be organised to derive themes. Data will be entered into NVivo.12 to support organisation and analysis. Three stages of thematic synthesis18 will be applied to data derived from all studies included in the systematic review.

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First stage: coding of text

Selected studies will be coded line by line by HaH, and results validated by PF. Any discrepancies in coding will be resolved through discussion or in further consultation with KL.

Second stage: developing descriptive themes

Reviewers will look for similarities and differences between the codes in order to start grouping them into a hierarchical tree structure[19]. On the basis of similarities and differences between the data, HaH will compile codes into descriptive topics. Data extracted from the first stage will be analysed to create potential themes and organised to form a hierarchical tree structure determined by themes. The aim of the hierarchical tree structure is to show themes with explanatory data and capture similarities within the data wherever possible19. The results for this stage will be discussed with PF with discrepancies in coding resolved through discussion and consultation with KL.

Third stage: generating analytical themes

At this stage, analytical themes will be developed with analysis extending beyond the initial data reported in the previous stages[20]. Analytical themes will be developed separately by HaH and PF and discussed, with KL included at this stage where consensus cannot be reached. Following thematic synthesis, results will be compared across studies to address review objectives and reach an understanding regarding the beliefs and attitudes towards mental health problems, acceptability of CBT and barriers to access. Comparison will be made regarding these beliefs and attitudes to examine any differences across Muslim communities.

Discussion

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This review will examine the beliefs and attitudes regarding mental health difficulties in adult Muslims, differences between communities towards the acceptability of CBT for the treatment of common mental health conditions and barriers to access evidence based psychological therapy. Identifying the attitudes of Muslim people toward mental health and treatment, especially CBT will help to understand barriers that prevent Muslims from accessing mental health services. Within the Improving Access to Psychological Therapies program[21]attempts to reduce barriers have predominantly focused on addressing provider and systemic barriers[12]. Much less focus however, has been given to individual barriers across cultures such as beliefs regarding mental health and attitudes towards psychological therapies[22]. This review has potential to understand individual level barriers and address current disparities in access to psychological therapies for different Muslim communities. Comparison across different Muslim communities will help to establish the degree of confidence in data synthesis to inform subsequent decision-making. Gaining a greater understanding regarding barriers within Muslim communities also offers potential to inform current disparities in access to psychological therapies for members of the wider Black Asian and Minority Ethnic (BAME) community[13].

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Award/Grant number is not applicable

Abbreviations

BAME Black Asian and Minority Ethnic.

CBT Cognitive Behaviour Therapy.

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MMAT Mixed Methods Appraisal Tool for the appraisal of qualitative, quantitative, and mixed methods research.

PRISMA-P Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols.

SPIDER Sample, Phenomenon of Interest, Design, Evaluation representing.

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3, 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4, 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	1
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4,5,6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6, 7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7, 8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8, 9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7,8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	N/A
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	N/A
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9, 10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	N/A
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Appendix 2: Ovid MEDLINE search strategy

Ovid MEDLINE(R) 1946 to December 17, 2019

1 Cognitive behavio#r therapy .af.

2 CBT .af.

3 Psychological therapy .af.

4 Psychotherapy .af.

5 (Cognitive behavio#r treatment or Psychological treatment or psychological intervention or Behavio#r intervention or psychological counsel#ing) .ti,ab.

6 1 OR 2 OR 3 OR 4 OR 5

7 Muslim* .af.

8 Islam* .af

9 7 or 8

10 Attitude* .af.

11 Belief* .af.

12 (Perception or Perspective* or Opinion* or Experience* or Thought*) .ti,ab.

13 Understand* .af

14 (Comprehension or Overview) .ti,ab.

15 10 OR 11 OR 12 OR 13 OR 14

16 barrier* .af.

17 Obstacle* or Challenge* or Difficult* or Issue* or Problem*) .ti,ab.

18 Treatment gap .af.

19 16 OR 17 OR 18

20 Depression .af.

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21	Depressive Disorder .af.
22	(depressed or melancholi* or dysthymi* or low mood) .ti,ab.
23	Anxiety .af.
24	Anxiety disorder .af.
25	(anxious or stress or worry) .ti,ab.
26	mental health .af.
27	(mental illness* or mental disorder* or psychiatric illness* or Psychiatric disorder*) .ti,ab.
28	20 or 21 or 22 or 23 or 24 or 25 or 26 or 27
29	6 and 9 and 15 and 19 and 28
30	29 (1980- 18/12/2019 + Human)

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Appendix 3: Saudi Digital Library (SLD) search strategy

1	العلاج المعرفي السلوكي TX
2	العلاج النفسي TX
3	(المعالجة النفسية or التدخل النفسي or الاستشارة النفسية) TI,AB
4	1 or 2 or 3
5	الاسلام TX
6	المسلمين TX
7	5 or 6
8	اتجاه TX
9	موقف TX
10	(مواقف or اتجاهات or اعتقاد or افكار or تصور or منظور) TI,AB
11	فهم TX
12	(توجه or رأي or وجهة نظر or استيعاب) TI, AB
13	8 or 9 or 10 or 11 or 12
14	صعوبات TX
15	(حواجز or معوقات or عوائق or تحديات or مشاكل) TI,AB
16	فجوه علاجيه TX
17	14 or 15 or 16
18	الاكتئاب TX
19	القلق TX
20	المرض النفسي TX
21	(الصحة النفسية or الامراض النفسية or الاضطرابات النفسية) TI,AB

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22	18 or 19 or 20 or 21
23	4 and 7 and 13 and 17 and 22
24	23 (1980-2019)

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Secondary Subject Heading:	Mental health
Keywords:	MENTAL HEALTH, Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Depression & mood disorders < PSYCHIATRY, Anxiety disorders < PSYCHIATRY

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Understanding the beliefs and attitudes towards mental health problems held by Muslim communities and acceptability of cognitive behavioural therapy as a treatment: systematic review and thematic synthesis protocol

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Authors Contributions HaH drafted the manuscript and led the design of the protocol and provided input across all areas. PF provided guidance during development of the protocol, provided advice across all areas and commented on all drafts. KL provided comments on full drafts.

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Abstract

Introduction Many barriers prevent Muslims' accessing mental health services, The aim of this systematic review is to gain an understanding regarding these barriers and consider how they vary across different Muslim communities resident in different countries.

Methods and Analysis Systematic review of PubMed/MEDLINE, CINAHL, PsycINFO, Ovid MEDLINE, Embase and Index Islamicus databases for studies published in English in addition to the Saudi Digital Library (SDL) for studies published in Arabic. The review will include quantitative, qualitative, and mixed methods studies published in peer-review journals since 1980. Methodological quality and risk of bias of included papers will be critically appraised independently according to the Mixed Methods Appraisal Tool (MMAT). Thematic synthesis will be used to extract outcome and analyse data from studies included in the review.

Ethics and Dissemination There are considered to be no ethical issues. Findings will be disseminated in both English and Arabic to clinicians and researchers via journal publication and conference presentation(s).

Trial Registration Number PROSPERO and registration number: CRD42020192854

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Strengths and limitations of this study

- First protocol for a systematic review and thematic synthesis examining barriers that prevent Muslims accessing mental health services.
- The review is inclusive regarding barriers held by different Muslim communities that live in different countries to facilitate comparisons.
- Studies restricted to those published in English or Arabic since 1980.
- Due to limited resources, in this review focus has only been on CBT.

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Introduction

Islam is the dominant religion in 56 countries around the globe and has more than 1.9 billion followers[1]. The majority of Muslims reside in the Islamic world that primarily covers the Middle East, Africa, and Asia. However, Muslims are also resident in many non-Muslim countries as a minority group having immigrated for economic, political or social reasons[2]. Muslims may be affected and differ by their residential and cultures norms[3]. For example, although the Qur'an clearly states that certain behaviours such as drinking alcohol and eating pork are forbidden, it has been reported that 25% of Korean Muslims drink alcohol with 16% eating pork[4]. Participants in this study reported that engaging in such behaviours arose as a response to their cultural upbringing, indicating that differences exist between Muslim communities residing in different countries.

Differences have also been reported in areas such as engaging with, and access to, mental health services. When accessing mental health services, Muslims experience a lower recovery rate (40.3%), compared to Christians (54.5%) and Jews (49.5%)[5, 6] Worse recovery rates for Muslims potentially reflecting a strong relationship between the Islamic religion with their response to all external events varying with the degree of religiosity[7]. Furthermore, compared to other religious groups, Muslims have also been reported to experience a range of additional barriers that prevent them from accessing mental health services. Within the UK, Muslims have been identified to be at a significant disadvantage when using mental health services[8] with access to psychological services remaining low in the UK[5].

Whilst many Muslims believe that mental health difficulties represent medical problems, some believe the aetiology of mental health difficulties is associated with beliefs in Evil Eye, possession, black magic[9] or as a punishment from Allah for sinful acts[10, 11, 12]. Additionally,

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limited access to mental health services for Muslims has also been reported[13], resulting in treatment sought from religious healers rather than mental health services[13, 14]. Whilst some barriers to access are known, little is known about how these differ across Muslim communities resident in different countries.

However, a systematic review conducted on Muslims and non-Muslims in Middle Eastern Arab countries has identified a range of barriers engaging with mental health interventions[15]. Barriers included the acceptance of interference in the cultural context (e.g., beliefs and values and the stigma of mental illnesses), community and system factors (e.g., access to services) and clinical participation processes (e.g., the expectations of the client)[15]. These barriers are similar to those reported in Western societies examining access to effective treatment for anxiety and depression. These included factors related to culture, the provider such as lack of knowledge of mental health problems and of a systemic nature such as the limited availability of specialty mental health providers[16]. Somewhat surprisingly however, individual barriers[16] relating to an understanding of mental health and treatment were not reported.

This systematic review seeks to gain a fuller understanding concerning barriers that prevent Muslims in different countries from accessing mental health service with a secondary focus examining barriers that prevent Muslims from accessing CBT as treatment. Examining barriers effecting Muslim communities resident in different countries will improve understanding regarding the extent to which any differences permit conclusions to be generalised across community.

Aims and Objectives

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Aim

Understand barriers that prevent adult Muslims from accessing mental health services with comparisons undertaken between Muslim communities resident in different countries.

Objectives:

Gain an understanding of barriers that prevent Muslim communities resident in different countries accessing mental health services.

Appreciate beliefs and attitudes regarding causes and treatment of mental health difficulties.

Evaluate the extent to which beliefs held towards mental health and treatment vary between Muslim communities resident in different countries.

Appreciate the extent that CBT may address barriers for adult Muslims.

Method

This systematic review protocol adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA) (see online supplementary appendix 1).

Eligibility Criteria:

Empirical studies published in peer-review journals and meeting review-specific eligibility criteria informed by the SPIDER search tool for use in systematic reviews of qualitative and mixed method research[17]. All studies will be published since 1980 in the English and Arabic language. Grey literature will be excluded.

Study Population:

The study population includes working age male and female adult Muslims that have either not received psychological therapies for the treatment of mental health difficulties or only

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received treatment with CBT. Working age varies between countries and as such will differ as an inclusion criterion. On occasions participant religion may not be specified in the description of the participants. In such cases, studies conducted in countries where the population is above 90% Muslim (e. g., Afghanistan, Iran, Libya, Saudi Arabia, Somalia, Turkey, Pakistan) will be included. Studies that focus on health care professionals, family or carers views will be excluded.

Interventions:

Studies focussing on general barriers to accessing mental health services will not be limited to any specific psychological intervention. However, studies focused on barriers to accessing a specific psychological therapy will be restricted to CBT[18].

Comparators:

Regardless of any comparator condition used in quantitative studies, all studies meeting inclusion criteria will be included.

Study Designs:

Qualitative, quantitative, questionnaire-based and mixed methods research designs will be included.

Outcomes:

Understand barriers that prevent Muslims from accessing mental health services and identification of barriers accessing CBT for mental health difficulties.

Language:

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Studies in the English or Arabic language.

Patient and Public Involvement:

No patients involved.

Information sources

In the English language the search will be based on the following research databases: PubMed/MEDLINE, CINAHL, PsycINFO, Ovid MEDLINE, Embase, and the Index Islamicus religious database. The Saudi Digital Library (SDL) electronic database will be searched for publications in Arabic.

Search Strategy

Truncation# or Wild Cards* will be used following appropriate search terms to support searches for variations on a word formed through different suffixes. For example, Depressi* to include depression or depressive disorder or depressive symptoms or major depressive disorder. Or # if the word has a different spelling such as behaviour#. Where available, search terms will be adjusted for use with specific bibliographic databases filters. Truncation or Wild Cards are not used within the SDL. Reference lists will be hand searched for all included studies to identify further studies (see online supplementary appendix 2, 3). HaH reviewed the research strategy with the University of Exeter Librarian competent in this area.

Study Selection

Prior to screening, a calibration exercise will be undertaken between HaH and PF. All titles and abstracts yielded by the search against inclusion/exclusion criteria will be screened by HaH. Full text of articles included from the title abstract screen will be independently

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reviewed by HaH and PF with discrepancies regarding inclusion resolved through discussion. Where consensus is not reached, differences will be resolved by wider discussion with KL for final decision making.

Data Extraction

Descriptive data regarding background characteristics –population, year of publication, country of study, study aim, study design, population, type of mental health difficulties if experienced– will be extracted from each study included in the review. Consistent with the aim of qualitative synthesis to extend beyond primary studies, additional data extracted will focus on the analysis and interpretations of the study authors towards themes, sub-themes, beliefs, models, attitudes and other relevant information associated with the study objectives. Data extraction forms will be developed and piloted a priori. For studies published in English, data extraction will be carried out by HaH with a random sample checked by PF. Data extraction for studies published in the Arabic language will only be undertaken by HaH.

Risk of Bias of Individual Studies

Methodological quality and risk of bias of included papers will be critically appraised independently by HaH and PF according to the Mixed Methods Appraisal Tool (MMAT)[19].

Data Analysis and Synthesis

Thematic synthesis[20] will be used to support interpretation of data obtained from qualitative, quantitative and mixed methods research beyond that undertaken by authors of included studies. Arising data will be organised to derive themes. Data will be entered into NVivo.12 to support organisation and analysis. Three stages of thematic synthesis[20] will be applied to data derived from all studies included in the systematic review.

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First stage: coding of text

Selected studies will be coded line by line by HaH, and results validated by PF. Any discrepancies in coding will be resolved through discussion or in further consultation with KL.

Second stage: developing descriptive themes

Reviewers will look for similarities and differences between codes in order to start grouping them into a hierarchical tree structure[21]. On the basis of similarities and differences between the data, HaH will compile codes into descriptive topics. Data extracted from the first stage will be analysed to create potential themes and organised to form a hierarchical tree structure determined by themes. The aim of the hierarchical tree structure is to show themes with explanatory data and capture similarities within the data wherever possible[21]. The results for this stage will be discussed with PF with discrepancies in coding resolved through discussion and consultation with KL.

Third stage: generating analytical themes

At this stage, analytical themes will be developed with analysis extending beyond the initial data reported in the previous stages[22]. Analytical themes will be developed separately by HaH and PF and discussed, with KL included at this stage where consensus cannot be reached. Following thematic synthesis, results will be compared across studies to address review objectives and reach an understanding regarding the barriers to access and acceptability of CBT to address these barriers. Where they arise, comparisons will be made regarding these barriers with respect to country of residence.

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Discussion

This systematic review will examine barriers that prevent adult Muslims from accessing mental health services. Within the Improving Access to Psychological Therapies (IAPT) program[23], attempts to reduce barriers have predominantly focused on addressing provider and systemic barriers[16] and restricted to services predominantly providing CBT. However, differences in barriers between Muslims resident in different countries will be explored to examine the potential that barriers experienced by the IAPT programme may vary by country of residence. Much less focus has been focussed on examining individual barriers across cultures such as beliefs regarding mental health and attitudes towards psychological therapies[24]. Comparison across Muslims resident in different countries will therefore help establish a degree of confidence in data synthesis to inform decision-making varying by country.

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Award/Grant number is not applicable

Competing interests there are no conflicts of interest

Abbreviations

CBT Cognitive Behaviour Therapy.

MMAT Mixed Methods Appraisal Tool for the appraisal of qualitative, quantitative, and mixed methods research.

PRISMA-P Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols.

SPIDER Sample, Phenomenon of Interest, Design, Evaluation representing.

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3, 4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4, 5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	1
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4,5,6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	6, 7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7, 8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	8, 9

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7,8
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	N/A
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	N/A
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9, 10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	N/A
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Appendix 2: Ovid MEDLINE search strategy

Ovid MEDLINE(R) 1946 to December 17, 2019

1 Cognitive behavio#r therapy .af.

2 CBT .af.

3 Psychological therapy .af.

4 Psychotherapy .af.

5 (Cognitive behavio#r treatment or Psychological treatment or psychological intervention or Behavio#r intervention or psychological counsel#ing) .ti,ab.

6 1 OR 2 OR 3 OR 4 OR 5

7 Muslim* .af.

8 Islam* .af

9 7 or 8

10 Attitude* .af.

11 Belief* .af.

12 (Perception or Perspective* or Opinion* or Experience* or Thought*) .ti,ab.

13 Understand* .af

14 (Comprehension or Overview) .ti,ab.

15 10 OR 11 OR 12 OR 13 OR 14

16 barrier* .af.

17 Obstacle* or Challenge* or Difficult* or Issue* or Problem*) .ti,ab.

18 Treatment gap .af.

19 16 OR 17 OR 18

20 Depression .af.

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21	Depressive Disorder .af.
22	(depressed or melancholi* or dysthymi* or low mood) .ti,ab.
23	Anxiety .af.
24	Anxiety disorder .af.
25	(anxious or stress or worry) .ti,ab.
26	mental health .af.
27	(mental illness* or mental disorder* or psychiatric illness* or Psychiatric disorder*) .ti,ab.
28	20 or 21 or 22 or 23 or 24 or 25 or 26 or 27
29	6 and 9 and 15 and 19 and 28
30	29 (1980- 18/12/2019 + Human)

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Appendix 3: Saudi Digital Library (SLD) search strategy

1	العلاج المعرفي السلوكي TX
2	العلاج النفسي TX
3	(المعالجة النفسية or التدخل النفسي or الاستشارة النفسية) TI,AB
4	1 or 2 or 3
5	الاسلام TX
6	المسلمين TX
7	5 or 6
8	اتجاه TX
9	موقف TX
10	(مواقف or اتجاهات or اعتقاد or افكار or تصور or منظور) TI,AB
11	فهم TX
12	(توجه or رأي or وجهة نظر or استيعاب) TI, AB
13	8 or 9 or 10 or 11 or 12
14	صعوبات TX
15	(حواجز or معوقات or عوائق or تحديات or مشاكل) TI,AB
16	فجوه علاجيه TX
17	14 or 15 or 16
18	الاكتئاب TX
19	القلق TX
20	المرض النفسي TX
21	(الصحة النفسية or الامراض النفسية or الاضطرابات النفسية) TI,AB

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22	18 or 19 or 20 or 21
23	4 and 7 and 13 and 17 and 22
24	23 (1980-2019)

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